

Auteur: Joe Ayoub
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$$\begin{aligned}\frac{1-3i}{1-i} + \frac{i}{2-i} &= \frac{(1-3i) \cdot (2-i) + (1-i) \cdot i}{(1-i) \cdot (2-i)} \\&= \frac{2+i-6i-3+i+1}{2-i-2i-1} \\&= \frac{-6i}{1-3i} = \frac{-6i \cdot (1+3i)}{1^2+3^2} \\&= \frac{18-6i}{10} = \frac{9-3i}{5}\end{aligned}$$

References